MAE 1001:
From numbers to loops: Unlocking Python's power

Introduction to Mechanical and Aerospace Engineering

Course website: https://gwu-mae1001.github.io/fall2026/

Prof. Kartik Bulusu, MAE Dept.

Graduate Teaching Assistant:
Preethi Siva Kumar (MAE Dept., MS student)

Learning Assistants:
Anya Spevacek (Senior, MAE Dept.)
Ergi Agaj (Senior, MAE Dept.)
Toby Pike (Junior, MAE Dept.)

GW School of Engineering & Applied Science
THE GEORGE WASHINGTON UNIVERSITY

Fall 2026

Photo: Kartik Bulusu



"Study hard what interests you the most in the most undisciplined, irreverent and original manner possible."

— Richard Feynmann

Computers calculate stuff



Icon Sources:

Floppy Disk by Diardha Gumi: https://thenounproject.com/browse/icons/term/floppy-disk/
harddisk by Start Up Graphic Design: https://thenounproject.com/browse/icons/term/harddisk/
calculations by Xinh Studio: https://thenounproject.com/browse/icons/term/calculations/
pi by Tom Fricker: https://thenounproject.com/browse/icons/term/pi/
calculations in physics by Olena Panasovska: https://thenounproject.com/browse/icons/term/calculations-in-physics/
waves by Joey Hiller: https://thenounproject.com/browse/icons/term/waves/
mockup chart by Dawid Sobolewski: https://thenounproject.com/browse/icons/term/mockup-chart/
mockup chart by Dawid Sobolewski: https://thenounproject.com/browse/icons/term/mockup-chart/

Computers need some language to communicate

English ?

Primitive constructs

- Words
  - Alphabets
- Sentences with words

Python ?

Primitive constructs

Numbers, strings

Basic math operators

+

-

*

/

> = <

School of Engineering & Applied Science
THE GEORGE WASHINGTON UNIVERSITY

GW

Prof. Kartik Bulusu, MAE Dept.

Fall 2026

MAE 1001 Introduction to Mechanical and Aerospace Engineering

Python programs

Sources:
What is computation? By Ana Bell: https://youtu.be/nykOeWgQcHM
Bilbo Baggins: https://en.wikipedia.org/wiki/Bilbo_Baggins
Bilbo Baggins: https://heroes-and-villain.fandom.com/wiki/Bilbo_Baggins

- Sequences of definitions and commands
- Everything in Python is an object
  - number 9, is an object
  - range() is an object
- Manipulate objects
- Objects:
  - have a type(), that determine what python program can do with them

Analogy

type



human

- Python has two kinds of objects
  - scalar: very basic and other objects can be made with them
  - non-scalar: have an internal structure

Source:
What is computation? By Ana Bell: https://youtu.be/nykOeWgQcHM

Our focus is on scalar objects

int	Represent integers including 0 (whole numbers)
float	Represent real numbers (with decimals)
bool	Represent Boolean values True and False
NoneType	Special and has one value, None

```
>>> type(9)
<class 'int'>

>>> type(9.0)
<class 'float'>

>>> type(range(9))
<class 'range'>
```

Integer & Float operators

i+j	Sum of integers
i-j	Difference
i*j	Product
i/j	Division
i%j	Remainder of i/j
ij**	i raised to the power j
i//j	Integer division

i and j are integers.
Result is an integer.

i and j are integers.
Results in decimal. (float)

i and j are integers.
Result is an integer.

Functions: A first look

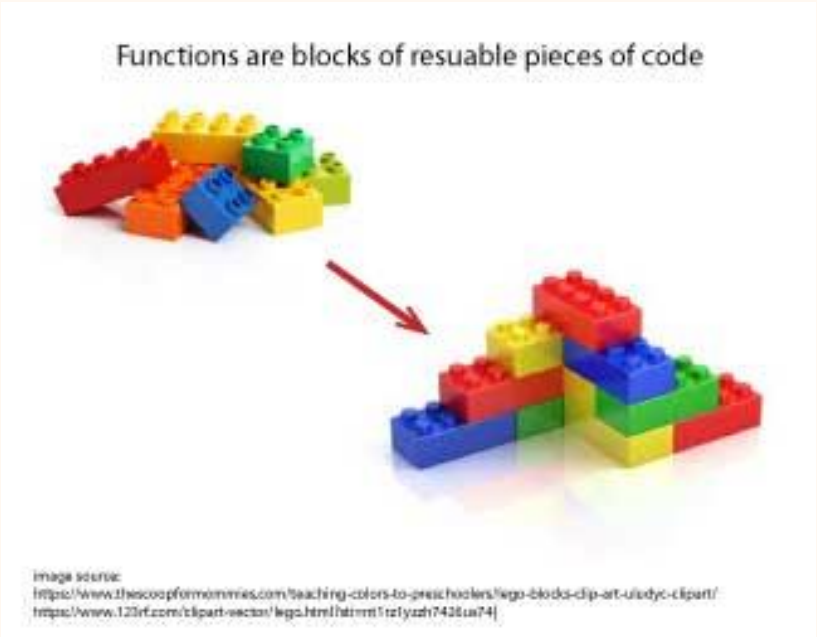
A function is group of related statements that performs a specific task.

functions

- Break your program into smaller and modular chunks.
- Make a program more manageable and modular as a program grows larger and larger
- Avoid repetition and makes the code reusable

print()

- Example of a function you just learned
- Built-in function in Python



print()

- By default, Python's print() function ends with a newline
- Commas between each entry outputs a space between each entry
- The entry of arguments of print() can be strings

More ways to use the print() function

0.1.2 - Strings

A *string* in Python is a sequence of letters, digits, or symbols (such as & or @) surrounded by either:

- A pair of double quotes, as in "Hello world!"
- A pair of single quotes, as in: 'Hello world!'

sep

```
print("Hello", "World!" ,"I", "love", Python", sep=",")
```

end

```
print("Hello", "World!" ,"I love Python", end=' ')
```

Demos

An escape character is a backslash \ followed by the character you want to insert.

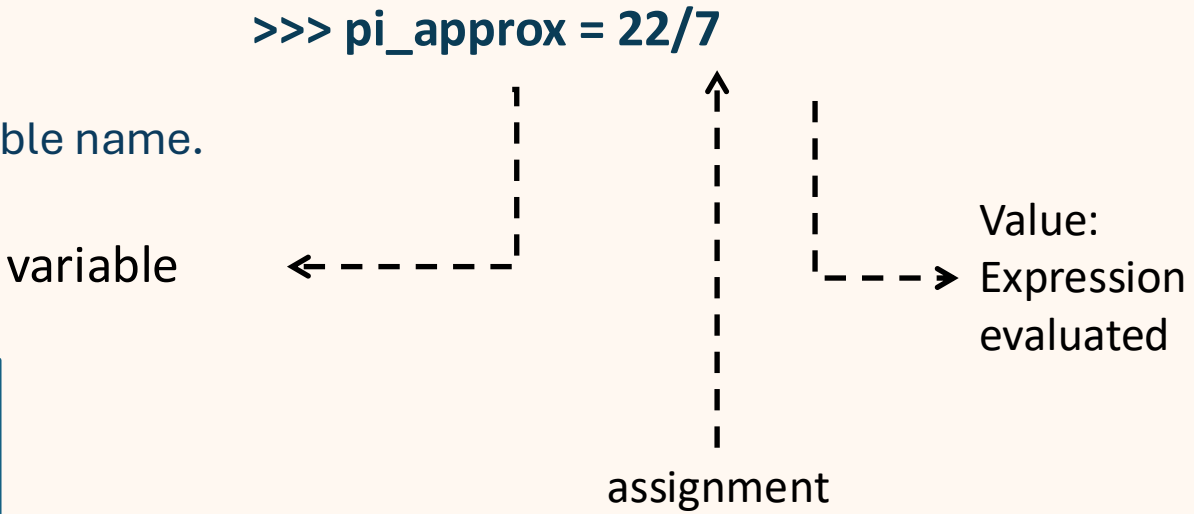
\'	Single Quote
\\	Backslash
\n	New Line
\r	Carriage Return
\t	Tab
\b	Backspace
\f	Form Feed
\ooo	Octal value
\xhh	Hex value

Source:
What is computation? By Ana Bell: https://youtu.be/nykOeWgQcHM

Variables

Give a name to the values of the expressions

The ***equal sign*** is an assignment of a value to a variable name.



- **Assignment binds the variable name to the value**
- Value is stored in the computer memory
- Invoke the variable name to retrieve the value
- **Variable refers to the value in the memory**

Built-in function range()

Python's range()function
- returns a sequence of numbers
- works only with integers

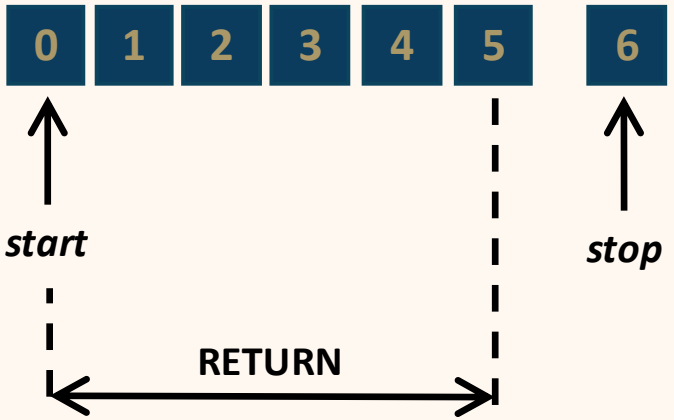
start:	at the value (default = 0)
step:	up or down at the increment value (default = 1)
stop:	at the value but not including it

range(stop)

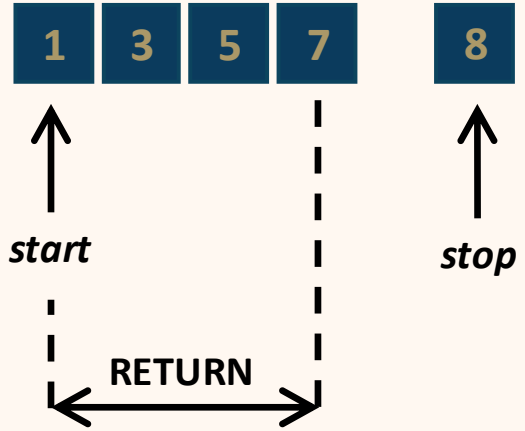
range(start, stop)

range(*start*, *stop*, *step*)

>>> range(6)



>>> range(1,8,2)



range()
- can be utilized in (for) loops
- to specify a range to iterate or do repetitions

Demos

Sources:
Codecademy: https://youtu.be/wxds6MAtUQ0

What is a ”loop” in computer programming ?

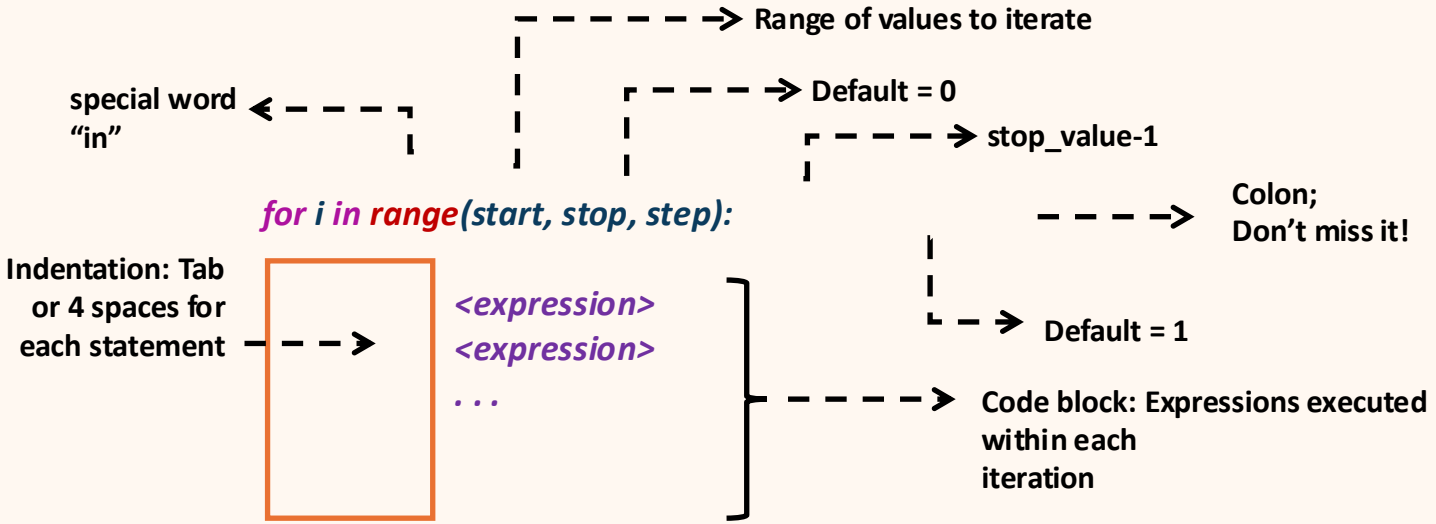


- Repeating routines
- From function to repeating functions
- Stopping condition

Skeleton of the for-loop

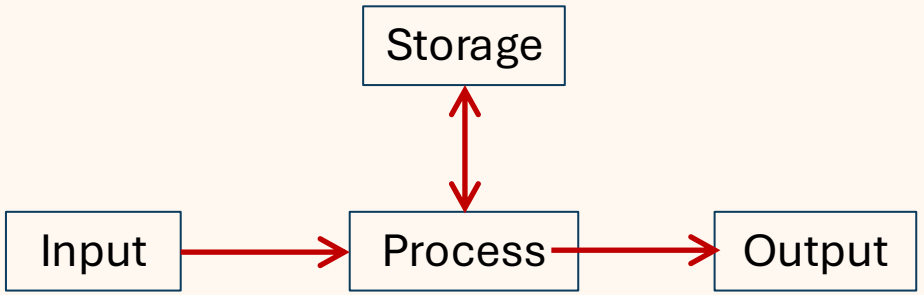
```
for i in range(start, stop, step):

        <expression>
        <expression>
        . . .
```

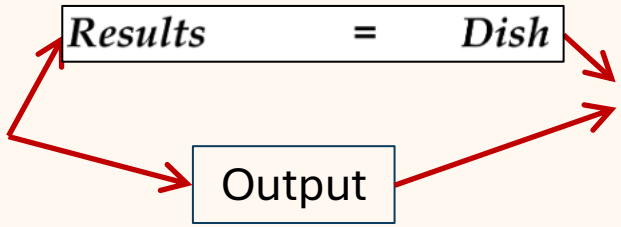


The Recipe Analogy

Program	=	*Recipe*
Laptop	=	*Cook*
Results (the program's output)	=	*Dish*



Counter-intuitively, we start looking at



print()

```
print("Hello World!")
print('Hello World!')
print('''George\nWashington was an
   amazing person. A university was
   named after him.''')
```